



# Girraween High School

## 2018

TRIAL HIGHER SCHOOL CERTIFICATE  
EXAMINATION

## Mathematics Extension 2

### General Instructions

- Reading time: 5 minutes
- Working time: 3 Hours
- Write using a black or blue pen
- Board approved calculators may be used
- Laminated reference sheets are provided
- Answer multiple choice questions by completely colouring in the appropriate circle on your multiple choice answer sheet on the front page of your answer booklet.
- In questions 11-16 start all questions on a separate page in your answer booklet and show all relevant mathematical reasoning and/or calculations.

**Total Marks: 100**

### **Section 1** (Pages 2– 5)      **10 Marks**

- Attempt Q1 - Q10
- Allow about 15 minutes for this section

### **Section 2** (Pages 5-13)      **90 marks**

- Attempt Q11 – Q16
- Allow about 2 hours and 45 minutes for this section

**Section 1** (10 marks)

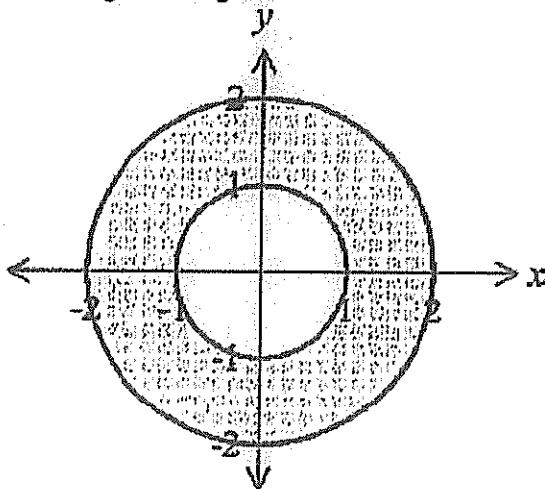
Allow about 15 minutes for this section.

Fill in the appropriate circle in your answer booklet.

1. Given that  $z = 1 + i$ , what is the value of  $z^8$ ?

- (A)  $-16$       (B)  $-8$       (C)  $8$       (D)  $16$

2. Consider the Argand diagram below.



Which inequality could define the shaded area?

- (A)  $0 \leq |z| \leq 2$   
(B)  $1 \leq |z| \leq 2$   
(C)  $0 \leq |z-1| \leq 2$   
(D)  $1 \leq |z-1| \leq 2$

3. The equation  $x^4 + px + q = 0$ , where  $p \neq 0$  and  $q \neq 0$  has roots  $\alpha, \beta, \gamma$  and  $\delta$ . What is the value of  $\alpha^4 + \beta^4 + \gamma^4 + \delta^4$ ?

- (A)  $-4q$       (B)  $p^2 - 2q$       (C)  $p^4 - 2q$       (D)  $p^4$

4. When  $x^y = e$  is implicitly differentiated with respect to  $x$ , the result for  $\frac{dy}{dx}$  is

- (A)  $\frac{-y}{x \log_e x}$     (B)  $\frac{y}{x \log_e x}$       (C)  $\frac{-x \log_e x}{y}$       (D)  $\frac{x \log_e x}{y}$

5. Which of the following is an expression for  $\int_0^{\frac{\pi}{2}} \frac{1}{1 + \sin x} dx$

after the substitution  $t = \tan \frac{x}{2}$  ?

(A)  $\int_0^1 \frac{1}{1 + 2t} dt$     (B)  $\int_0^1 \frac{2}{1 + 2t} dt$     (C)  $\int_0^1 \frac{1}{(1 + t)^2} dt$     (D)  $\int_0^1 \frac{2}{(1 + t)^2} dt$

6. What are the equations of the directrices of the hyperbola with equation

$$\frac{x^2}{144} - \frac{y^2}{25} = 1 ?$$

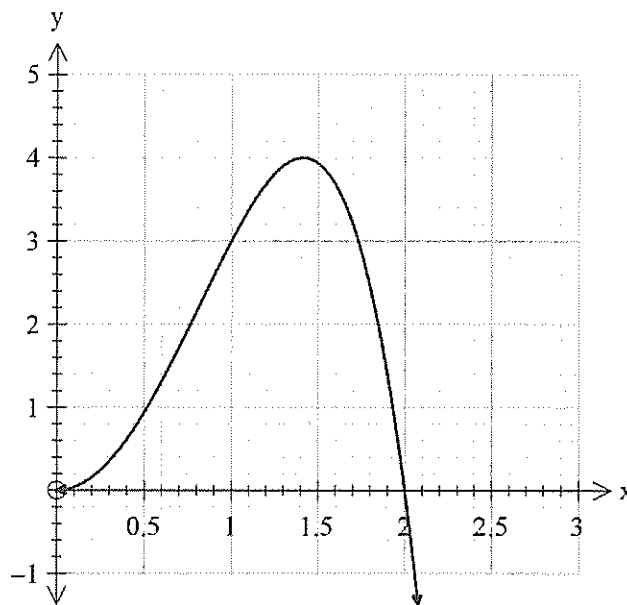
(A)  $x = \pm \frac{13}{144}$     (B)  $x = \pm \frac{13}{25}$     (C)  $x = \pm \frac{25}{13}$     (D)  $x = \pm \frac{144}{13}$

7. The region enclosed by  $y = \sin x$ ,  $y = 0$  and  $x = \frac{\pi}{2}$  is rotated about the  $y$ -axis

to produce a solid. What is the volume of this solid using the method of cylindrical shells?

(A)  $\pi$  cubic units    (B)  $\frac{\pi}{2}$  cubic units    (C)  $\frac{3\pi}{2}$  cubic units    (D)  $2\pi$  cubic units

8. The graph of  $y = 4x^2 - x^4$  is given below.



The region in the first quadrant bounded by the curve  $y = 4x^2 - x^4$  and the  $x$ -axis between  $x = 0$  and  $x = 2$  is rotated about the  $y$ -axis.

Which of the following is an expression for the volume,  $V$ , of the solid formed?

(A)  $V = 2\pi \int_0^4 \sqrt{4-y} \, dy$

(B)  $V = 4\pi \int_0^4 \sqrt{4-y} \, dy$

(C)  $V = 8\pi \int_0^4 \sqrt{4-y} \, dy$

(D)  $V = 16\pi \int_0^4 \sqrt{4-y} \, dy$

9. A wheel of radius 2 metres rotates at 1200 revolutions per minute.

What is the tangential velocity of a point on the wheel?

(A) 40 m/s

(B) 80 m/s

(C) 251 m/s

(D) 260 m/s

10. A particle of mass  $m$  is moving horizontally in a straight line. Its motion is opposed by a force of magnitude  $2m(v + v^2)$  Newtons when its speed is  $v$  m/s. At time  $t$  seconds the particle has a displacement of  $x$  metres from a fixed point  $O$  on the line and velocity  $v$  m/s.  
Which of the following is an expression for  $x$  in terms of  $v$ ?

(A)  $-\frac{1}{2} \int \frac{1}{1+v} dv$

(B)  $-\frac{1}{2} \int \frac{1}{v(1+v)} dv$

(C)  $\frac{1}{2} \int \frac{1}{1+v} dv$

(D)  $\frac{1}{2} \int \frac{1}{v(1+v)} dv$

## Section 2

### Question 11 (15 marks)

- a.  $z = p + 2i$ , where  $p$  is a real number, and  $w = 1 - 2i$  represent two complex numbers.

(i) Find  $\frac{z}{w}$  in the form  $a + ib$ , where  $a$  and  $b$  are real numbers. [2]

(ii) Given that  $\left| \frac{z}{w} \right| = 13$ , find all possible values of  $p$ . [2]

b.  $z = 1 - \sqrt{3}i$

(i) Find the values of  $|z|$  and  $\arg z$ . [2]

(ii) Find the exact value of  $z^6$ . [2]

c. (i) On an Argand diagram, sketch the locus of  $z$  represented by  $|z - 3| = 3$ . [2]

(ii) Explain why  $\arg(z - 3) = 2 \arg z$ . [2]

d. If  $2 + i$  is a root of  $P(x) = x^4 - 6x^3 + 9x^2 + 6x - 20$ , resolve  $P(x)$  into irreducible factors over the complex field. [3]

**Question 12** (15 marks)

a. Find

(i)  $\int x e^{-x} dx$  [2]

(ii)  $\int_0^{\frac{\pi}{2}} \frac{1}{2 - \cos x + 2 \sin x} dx$  [3]

(iii)  $\int_0^{\frac{\pi}{6}} \sec^3 2\theta d\theta$  [3]

b. (i) Find real numbers  $a, b, c$  and  $d$  such that:

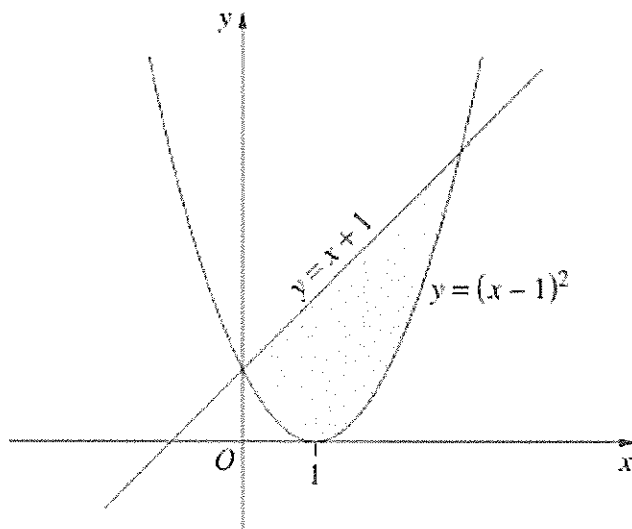
$$\frac{5x^3 - 3x^2 + 2x - 1}{x^2(x^2 + 1)} = \frac{a}{x} + \frac{b}{x^2} + \frac{cx + d}{x^2 + 1}$$
 [2]

(ii) Hence find  $\int \frac{5x^3 - 3x^2 + 2x - 1}{x^2(x^2 + 1)} dx$  [2]

c. The diagram shows the region enclosed by the curves  $y = x + 1$  and  $y = (x - 1)^2$ .

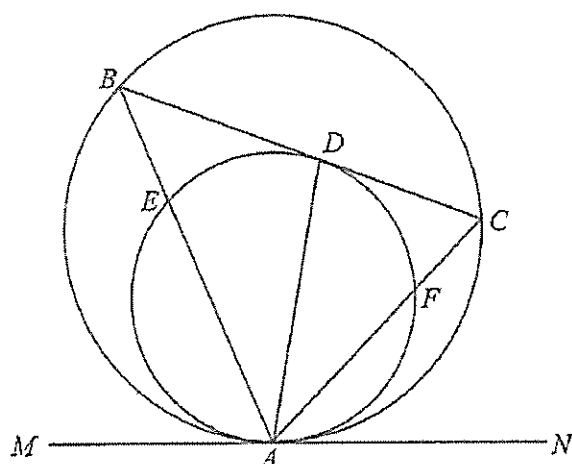
The region is rotated about the  $y$ -axis.

Find the volume of the solid using the method of cylindrical shells. [3]



**Question 13 (15 marks)**

a.



In the diagram, MAN is the common tangent to two circles touching internally at A.

B and C are two points on the larger circle such that BC is a tangent to the smaller circle with point of contact at D. AB and AC cut the smaller circle at E and F respectively.

(i) Copy or trace the into your answer booklet.

(ii) Show that  $AD$  bisects  $\angle BAC$  [4]

b. An ellipse has the equation  $\frac{x^2}{100} + \frac{y^2}{75} = 1$ .

(i) Sketch the curve, showing the coordinates of the foci and the equations of the directrices. [2]

(ii) Find the equation of the normal to the ellipse at the point  $P\left(5, 7\frac{1}{2}\right)$ . [2]

(iii) Find the equation of the circle that is tangential to the ellipse at  $P$  and  $Q\left(5, -7\frac{1}{2}\right)$ . [3]

c. (i) Show that the tangent to the curve  $xy = c^2$  at  $T\left(ct, \frac{c}{t}\right)$  is given by

$$x + t^2y = 2ct \quad [2]$$

(ii) The tangent cuts the  $x$  and  $y$  axes at  $A$  and  $B$  respectively.

Prove that  $T$  is the centre of the circle that passes through  $O$ ,  $A$  and  $B$

where  $O$  is the origin. [2]

**Question 14 (15 marks)**

a. (i) Prove that  $\int_0^a f(x) dx = \int_0^a f(a-x) dx$ . [2]

(ii) Hence, find the value of  $\int_0^2 x(2-x)^5 dx$ . [2]

b. Given that  $I_{2n+1} = \int_0^1 x^{2n+1} e^{x^2} dx$ , where  $n$  is a positive integer,

(i) Show that  $I_{2n+1} = \frac{1}{2} e - n I_{2n-1}$ . [3]

(ii) Hence, or otherwise, evaluate  $\int_0^1 x^5 e^{x^2} dx$ . [3]

c. (i) Use the binomial theorem to expand  $(\cos \theta + i \sin \theta)^3$ . [1]

(ii) Use De Moivre's Theorem and your result from (i) to prove that

$$\cos^3 \theta = \frac{1}{4} \cos 3\theta + \frac{3}{4} \cos \theta. \quad [2]$$

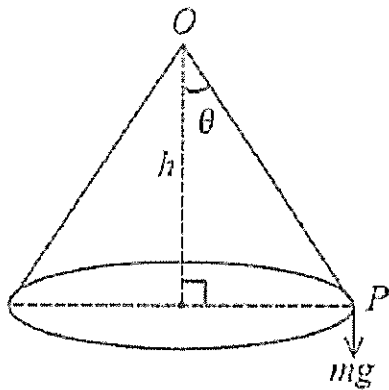
(iii) Hence, or otherwise, find the smallest positive solution of

$$4\cos^3 \theta - 3\cos \theta = 1 \quad [2]$$

**Question 15** (15 marks)

a. A mass of  $m$  kg at  $P$  is suspended by a light inextensible string from point  $O$ .

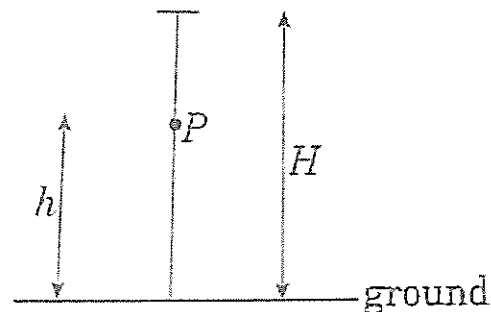
It describes a circle with a constant speed in a horizontal plane whose vertical distance below  $O$  is  $h$  metres.



(i) Show that  $\omega = \sqrt{\frac{g}{h}}$ . [2]

(ii) What is the period of the motion? [1]

b.



From a point on the ground an object of mass  $m$  kg is projected vertically upward with an initial speed of  $u$ . The object reaches a maximum height of  $H$  before falling back to the ground. The resistance to motion is equal to  $mkv^2$  and  $g$  is the acceleration due to gravity.

(i) Show that  $H = \frac{1}{2k} \ln \left( \frac{g + ku^2}{g} \right)$ . [2]

(ii)  $P$  is a point at height  $h$  above the point of projection.

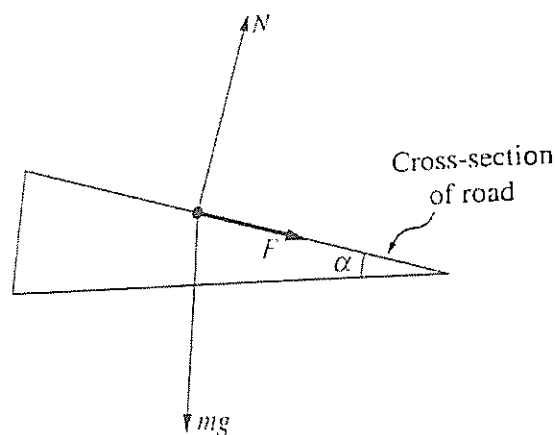
Let  $V$  be the speed of the object at  $P$  on its upward path when  $x = h$ .

Show that  $h = \frac{1}{2k} \ln \left( \frac{g + ku^2}{g + kV^2} \right)$ . [2]

(iii) During the downward path of the object it passes through  $P$  with half the speed of when it was first at  $P$ .

Show that  $V = \sqrt{\frac{3g}{k}}$ . [3]

c.



A road contains a bend that is part of a circle of radius,  $r$ . At the bend, the road is banked at an angle of  $\alpha$  to the horizontal. A car travels around the bend at constant speed,  $v$ . Assume that the car is represented by a point of mass  $m$ , and that the forces acting on the car are the gravitational force  $mg$ , a sideways frictional force  $F$  (acting down the road) and a normal reaction  $N$  to the road.

(i) By resolving the horizontal and vertical components of force, find expressions for

$$F \cos \alpha \text{ and } F \sin \alpha . \quad [2]$$

(ii) Show that  $F = \frac{m(v^2 - gr \tan \alpha)}{r} \cos \alpha . \quad [2]$

(iii) Suppose that the radius of the bend is 200 metres and that the road is banked to allow cars to travel at 100 km/h with no sideways friction force. Take  $g = 9.8 \text{ ms}^{-2}$ . Find the value of  $\alpha$ . [1]

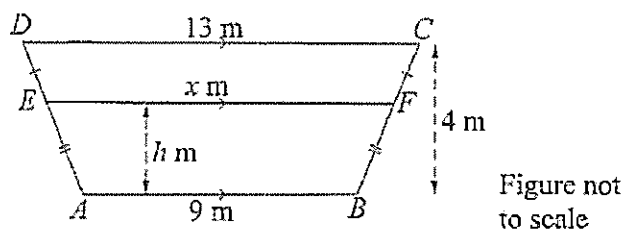
**Question 16 (15 marks)**

- a.  $P(x)$  is a polynomial of degree 5 such that  $P(x) - 1$  is divisible by  $(x - 1)^3$  and  $P(x)$  itself is divisible by  $x^3$ . Derive an expression for  $P(x)$ . [3]

- b. (i) The diagram below shows a trapezium  $ABCD$  whose parallel sides  $AB$  and  $DC$  are 9 metres and 13 metres respectively.

The distance between these sides is 4 metres and  $AD = BC$ .

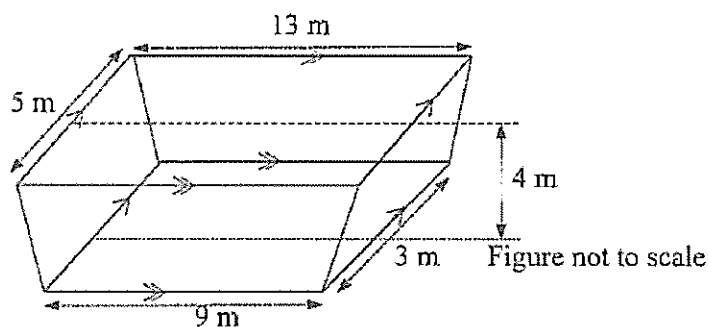
$EF$  is parallel to  $AB$  and the distance between them is  $h$  metres.



Show that  $EF = (9 + h)$  metres.

[2]

- (ii) The trench in the diagram below has a rectangular base with sides 9 metres and 3 metres. Its top is also rectangular with dimensions 13 metres and 5 metres. The trench has a depth of 4 metres and each of its four side faces is an isosceles trapezium.



Find the volume of the trench.

[4]

**Question 16 continues on Page 13**

- c. (i) Show that if  $y = mx + k$  is a tangent to the hyperbola  $xy = c^2$ ,  
then  $k^2 + 4mc^2 = 0$ . [3]
- (ii) Hence, find the equations of the tangents from the point  $(-1, -3)$   
to the rectangular hyperbola  $xy = 4$  and find their points of contact. [3]

*End of Examination*



①

GHS 2018 TRIAL NSC MATHEMATICS EXT. 2 SOLUTIONS  
MC.

1.  $z = 1 + i$ ,  $z^8 = ?$   
 $= \sqrt{2} \operatorname{cis} \frac{\pi}{4}$

$$\begin{aligned} z^8 &= (\sqrt{2} \operatorname{cis} \frac{\pi}{4})^8 \\ &= 2^4 (\cos 2\pi + i \sin 2\pi) \\ &= 16 \end{aligned}$$

**D**

2.  $1 \leq |z| \leq 2$

**B**

3.  $x^4 + px + q = 0$

$$\alpha^4 + p\alpha + q = 0$$

$$\beta^4 + p\beta + q = 0$$

$$\gamma^4 + p\gamma + q = 0$$

$$\delta^4 + p\delta + q = 0$$

$$\alpha^4 + \beta^4 + \gamma^4 + \delta^4 = -4p(\alpha + \beta + \gamma + \delta) - 4q$$

$$= -4p(0) - 4q$$

$$= -4q$$

**A**

4.  $x^y = e$

$$\log x^y = \log e$$

$$y \log x = 1$$

$$y \cdot \frac{1}{x} + \log x \frac{dy}{dx} = 0$$

$$\frac{y}{x} + \log x \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{-y}{x \log x}$$

**A**

①



(2)

$$5. \int_0^{\pi/2} \frac{1}{1 + \sin x} dx$$

$$t = \tan \frac{x}{2}$$

$$= \int_0^1 \frac{1}{1 + \frac{2t}{1+t^2}} \cdot \frac{2dt}{1+t^2}$$

$$\frac{dt}{dx} = \frac{1}{2} \sec^2 \frac{x}{2}$$

$$= \int_0^1 \frac{2}{(1+t^2+2t)} dt$$

$$dx = \frac{2dt}{\sec^2 \frac{x}{2}}$$

$$= \int_0^1 \frac{2}{(1+t)^2} dt$$

$$dx = \frac{2dt}{1+t^2}$$

$$\text{When } x = \frac{\pi}{2}, t = 1$$

$$\text{When } x = 0, t = 0$$

**D**

$$6. \frac{x^2}{144} - \frac{y^2}{25} = 1$$

$$b^2 = a^2(e^2 - 1)$$

$$e^2 - 1 = \frac{25}{144}$$

$$e^2 = \frac{169}{144}$$

$$e = \frac{13}{12}$$

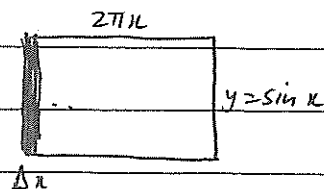
$$\text{Directrices : } x = \pm \frac{a}{e}$$

$$= \pm \frac{144}{13}$$

**D**

$$7. V = \lim_{\Delta x \rightarrow 0} \sum_{x=0}^{\pi/2} 2\pi x \sin x \Delta x$$

$$= 2\pi \int_0^{\pi/2} x \sin x dx$$



$$= 2\pi \left[ (x \cos x)^{\pi/2}_0 + \int_0^{\pi/2} \cos x dx \right]$$

$$= 2\pi \left[ \sin x \right]_0^{\pi/2}$$

$$= 2\pi$$

**D**



(3)

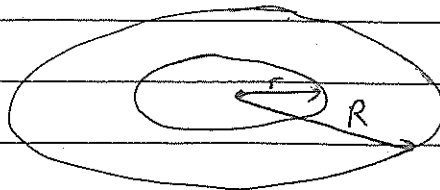
8.  $y = 4x^2 - x^4$

$$(x^4 - 4x^2 + 4) = -y + 4$$

$$(x^2 - 2)^2 = 4 - y$$

$$x^2 - 2 = \pm \sqrt{4 - y}$$

$$x^2 = 2 + \sqrt{4 - y} = R^2; \quad x^2 = 2 - \sqrt{4 - y} = r^2$$



$$A = \pi(R^2 - r^2)$$

$$= \pi(2 + \sqrt{4 - y} - (2 - \sqrt{4 - y}))$$

$$= \pi(2\sqrt{4 - y})$$

$$= 2\pi\sqrt{4 - y}$$

$$\Delta V = 2\pi\sqrt{4 - y} \Delta y$$

$$V = \lim_{\Delta y \rightarrow 0} \sum_{y \rightarrow 0}^4 2\pi\sqrt{4 - y} \Delta y$$

$$V = 2\pi \int_0^4 \sqrt{4 - y} dy$$

**A**

9.  $\omega = 1200 \text{ rpm}$

$$= \frac{1200 \times 2\pi}{60}$$

$$= 40\pi \text{ radians/sec}$$

$$v = r\omega$$

$$= 2 \times 40\pi$$

$$= 80\pi$$

$$\approx 251 \text{ m/s}$$

**C**

10.  $m\ddot{x} = -2m(v + v^2)$

$$\ddot{x} = -2(v + v^2)$$

$$v \frac{dv}{dx} = -2(v + v^2)$$

$$\frac{dx}{dv} = \frac{-1}{2(1+v)}$$

$$x = -\frac{1}{2} \int \frac{1}{1+v} dv$$

**A**



(4)

Question 11

a)  $z = p + 2i$  ;  $w = 1 - 2i$

i)  $\frac{z}{w} = \frac{p+2i}{1-2i} \times \frac{1+2i}{1+2i}$

$$= \frac{p + 2pi + 2i - 4}{1+4}$$

$$= \frac{p-4}{5} + \frac{2(p+1)}{5} i$$

(2)

ii)  $\left| \frac{z}{w} \right| = 13$

$$\frac{|z|}{|w|} = 13$$

$$\frac{p^2+4}{5} = 169$$

$$p^2 = 841$$

$$p = \pm \sqrt{841} = \pm 29$$

(2)

b)  $z = 1 - \sqrt{3}i$

i)  $|z| = \sqrt{1+3} = 2$

$$\arg z = -\frac{\pi}{3}$$

(2)

ii)  $z = 2 \operatorname{cis} -\frac{\pi}{3}$

$$z^6 = \left( 2 \operatorname{cis} -\frac{\pi}{3} \right)^6$$

$$= 2^6 (\cos -2\pi + i \sin -2\pi)$$

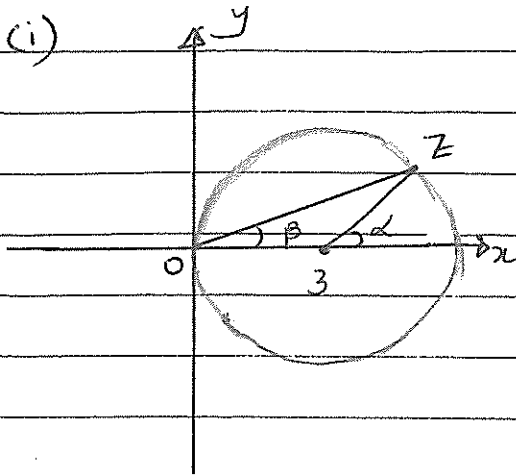
$$= 64$$

(2)



(5)

C. (i)



(2)

$$\therefore \alpha = \arg(z-3) \quad ; \quad \beta = \arg z$$

$\alpha = 2\beta$  ( $\angle$  at the centre is twice the  $\angle$  at the circumference subtended on same arc)

$$\therefore \arg(z-3) = 2\arg z$$

(2)

d.  $2+i$  is a root

$\therefore 2-i$  is also a root ( $P(x)$  has real coefficients)

$\therefore x^2 - 4x + 5$  is a factor.

$$\begin{array}{r}
 x^2 - 2x - 4 \\
 x^2 - 4x + 5 \overline{) x^4 - 6x^3 + 9x^2 + 6x - 20} \\
 \underline{x^4 - 4x^3 + 5x^2} \phantom{- 20} \\
 -2x^3 + 4x^2 + 6x \phantom{- 20} \\
 \underline{-2x^3 + 8x^2 - 10x} \phantom{- 20} \\
 -4x^2 + 16x - 20 \\
 \underline{-4x^2 + 16x - 20} \\
 0
 \end{array}$$

$$P(x) = (x^2 - 4x + 5)(x^2 - 2x - 4)$$

Roots of  $P(x)$  are  $2+i, 2-i, 1+\sqrt{5}, 1-\sqrt{5}$

Factors :  $(x - (2+i)), (x - (2-i)), (x - (1+\sqrt{5})), (x - (1-\sqrt{5}))$

(3)



(6)

## Question 12

a) i)

$$\int x e^{-x} dx$$

$$u = x \quad v = -e^{-x}$$

$$u' = 1 \quad v' = e^{-x}$$

$$= -x e^{-x} - \int -e^{-x} dx$$

$$= -x e^{-x} + \int e^{-x} dx$$

$$= -x e^{-x} - e^{-x} + c$$

(2)

ii)

$$\int_0^{\pi/2} \frac{1}{2 - \cos x + 2 \sin x} dx$$

$$\text{let } t = \tan \frac{x}{2}$$

$$dt = \frac{1}{2} \sec^2 \frac{x}{2} dx$$

$$= \int \frac{1}{2 - \left(\frac{1-t^2}{1+t^2}\right) + 2\left(\frac{2t}{1+t^2}\right)} \cdot \frac{2dt}{1+t^2}$$

$$= \frac{1}{2} (1+t^2) dx$$

$$dx = \frac{2dt}{1+t^2}$$

$$= \int_0^1 \frac{2 dt}{2(1+t^2) - (1-t^2) + 4t}$$

$$\text{When } x=0, t=0$$

$$x = \pi/2 \quad t=1$$

$$= \int_0^1 \frac{2}{1 + 3t^2 + 4t} dt$$

$$= \int_0^1 \frac{2}{(3t+1)(t+1)} dt$$

$$\frac{A}{3t+1} + \frac{B}{t+1} = \frac{2}{(3t+1)(t+1)}$$

$$= \int_0^1 \left( \frac{3}{3t+1} - \frac{1}{t+1} \right) dt$$

$$A(t+1) + B(3t+1) = 2$$

$$\text{Substitute } t = -1$$

$$-2B = 2 \Rightarrow B = -1$$

$$\text{Substitute } t = -\frac{1}{3}$$

$$\frac{2}{3} A = 2$$

$$A = 3$$

$$= \left[ \log(3t+1) - \log(t+1) \right]_0^1$$

$$= (\log 4 - \log 2) - (\log 1 - \log 1)$$

$$= \log 2$$

(3)



(7)

$$\text{iii) } \int_0^{\pi/6} \sec^3 2\theta \, d\theta$$

$$= \int_0^{\pi/6} \sec^2 2\theta \cdot \sec 2\theta \, d\theta$$

$$= \left[ \frac{\tan 2\theta \cdot \sec 2\theta}{2} \right]_0^{\pi/6} - \int_0^{\pi/6} \sec 2\theta \tan^2 2\theta \, d\theta$$

$$= \left( \frac{\tan \frac{\pi}{3} \cdot \sec \frac{\pi}{3}}{2} \right) - \int_0^{\pi/6} \sec 2\theta (\sec^2 2\theta - 1) \, d\theta$$

$$= \frac{\sqrt{3}}{2} \cdot 2 - \int_0^{\pi/6} (\sec^3 2\theta + \sec 2\theta) \, d\theta$$

$$= \sqrt{3} - \int_0^{\pi/6} \sec^3 2\theta \, d\theta + \int_0^{\pi/6} \sec 2\theta \, d\theta$$

$$2 \int_0^{\pi/6} \sec^3 2\theta \, d\theta = \sqrt{3} + \int_0^{\pi/6} \sec 2\theta \, d\theta$$

$$= \sqrt{3} + \int_0^{\pi/6} \frac{\sec 2\theta (\sec 2\theta + \tan 2\theta)}{(\sec 2\theta + \tan 2\theta)} \, d\theta$$

$$= \sqrt{3} + \frac{1}{2} \int_0^{\pi/6} \frac{2 \sec^2 2\theta + 2 \sec 2\theta \tan 2\theta}{\sec 2\theta + \tan 2\theta} \, d\theta$$

$$= \sqrt{3} + \frac{1}{2} \ln [\sec 2\theta + \tan 2\theta]_0^{\pi/6}$$

$$= \sqrt{3} + \frac{1}{2} \ln (2 + \sqrt{3})$$

$$\therefore \int_0^{\pi/6} \sec^3 2\theta \, d\theta = \frac{\sqrt{3}}{2} + \frac{1}{4} \ln (2 + \sqrt{3})$$

$$\text{let } u = \sec 2\theta$$

$$= \frac{1}{\cos 2\theta}$$

$$= (\cos 2\theta)^{-1}$$

$$u' = -1 (\cos 2\theta)^{-2} \cdot -2 \sin 2\theta$$

$$= \frac{2 \sin 2\theta}{\cos^2 2\theta}$$

$$= 2 \sec 2\theta \cdot \tan 2\theta$$

$$v' = \sec^2 2\theta$$

$$v = \frac{\tan 2\theta}{2}$$

(3)



(8)

$$b) i) \frac{5x^3 - 3x^2 + 2x - 1}{x^2(x^2 + 1)} = \frac{a}{x} + \frac{b}{x^2} + \frac{cx + d}{x^2 + 1}$$

$$5x^3 - 3x^2 + 2x - 1 = ax(x^2 + 1) + b(x^2 + 1) + (cx + d)x^2$$

$$= (a+c)x^3 + (b+d)x^2 + ax + b$$

comparing :

Constant term  $\Rightarrow b = -1$

Coefficients of  $x$  :  $a = 2$

Coefficients of  $x^2$  :  $b + d = -3$

$$d = -2$$

Coefficients of  $x^3$  :  $a + c = 5$

$$c = 3$$

$$\therefore a = 2, b = -1, c = 3, d = -2$$

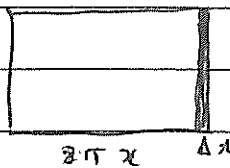
(2)

$$ii) \int \frac{5x^3 - 3x^2 + 2x - 1}{x^2(x^2 + 1)} dx = \int \left( \frac{2}{x} - \frac{1}{x^2} + \frac{3x - 2}{x^2 + 1} \right) dx$$

$$= \int \left( \frac{2}{x} - \frac{1}{x^2} + \frac{3x}{x^2 + 1} - \frac{2}{x^2 + 1} \right) dx$$

$$= 2 \log x + \frac{1}{x} + \frac{3}{2} \log(x^2 + 1) - 2 \tan^{-1} x + C$$

c)



$$y_2 - y_1 =$$

$$(x+1) - (x-1)^2$$

$$= -x^2 + 3x$$

$$\Delta V = 2\pi x (-x^2 + 3x) \Delta x$$

$$V = \lim_{\Delta x \rightarrow 0} \sum_{x=0}^3 2\pi (-x^3 + 3x^2) \Delta x$$

$$= 2\pi \int_0^3 (-x^3 + 3x^2) dx$$

$$= 2\pi \left[ -\frac{x^4}{4} + x^3 \right]_0^3$$

$$= \frac{27\pi}{2}$$

Points of intersection :

$$(x-1)^2 = x+1$$

$$x^2 - 2x + 1 = x + 1$$

$$x^2 - 3x = 0$$

$$x = 0, 3$$

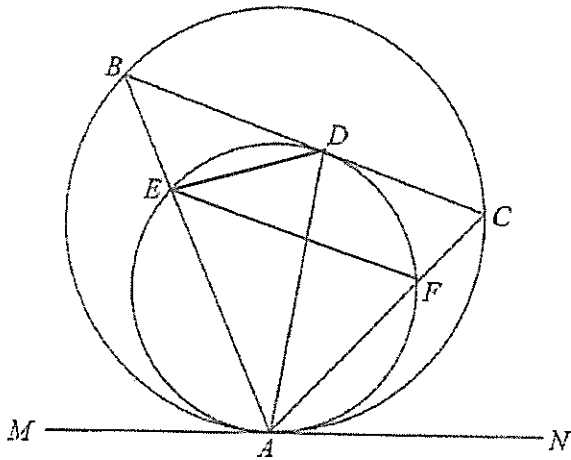
(3)



9

### Question 13

|    |    |
|----|----|
| 4) | i) |
|----|----|



(11) Construct EF, ED.

$\angle ABC = \angle NAC$  ( $\angle$  in the alternate segment)

$$\angle AEF = \angle NAC \quad (\angle \text{ in the alternate segment})$$

$\therefore EF \parallel BC$  (corresponding  $\angle$ s on transversal  $AB$ )

$$\angle DEF = \angle BDE \text{ (alternate } \angle s, EF \parallel BC)$$

$\angle DEF = \angle DAF$  ( $\angle$ s subtended by arc DF at the circumference of circle AEF)

$\angle BDE = \angle DAE$  ( $\angle$  in alternate segment in circle AEF)

$$\therefore \angle DAF = \angle DAE$$

$\therefore AD$  bisects  $\angle BAC$

$$b) \frac{26^2}{100} + \frac{4^2}{75} = 1$$

$$\underline{a=10, \quad b=5\sqrt{3}}$$

i)  $b^2 = a^2(1 - e^2)$

$$75 = 100(1 - e^2)$$

$$e^2 = \frac{1}{4}$$

$$e = \frac{1}{2}$$

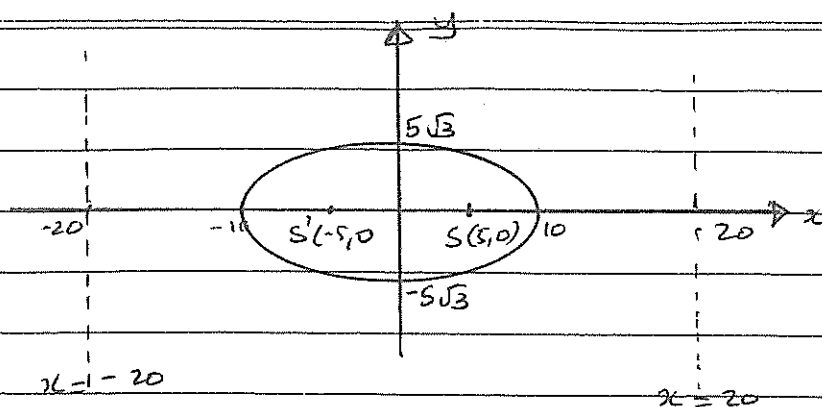
Foci :  $(\pm ae, 0)$

Foci :  $(\pm 5, 0)$  ; Directrices :  $x = \pm \frac{a}{e}$

Directions :  $x = \pm 20$



(16)



$$ii) \frac{x^2}{100} + \frac{y^2}{75} = 1$$

Differentiating implicitly,

$$\frac{2x}{100} + \frac{2y}{75} \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{3x}{4y}$$

At  $(5, 7\frac{1}{2})$

$$m_{\text{tangent}} = \frac{-3(5)}{4(7.5)} = -\frac{1}{2}$$

$$\therefore m_{\text{normal}} = 2$$

$$E_{\text{normal}} : y - 7.5 = 2(x - 5)$$

$$y = 2x - 2.5$$

iii) Circle and ellipse have common tangent at P

$\therefore$  Normal at P also at right  $\angle$ s to circle at P

$\therefore$  Normal at P passes through centre, C, of circle.

$$\therefore C \text{ lies on } y = 2x - 2.5 \quad \text{--- (1)}$$

$$\text{Tangent at Q } (5, -7.5) \text{ has gradient, } m = \frac{-3(5)}{4(-7.5)} = \frac{1}{2}$$

$$\therefore m_{\text{normal at Q}} = -2$$

$$E_{\text{normal at Q}} : y = -2x + 2.5$$

$\Rightarrow$  also passes through centre, C, of circle

$$\therefore C \text{ lies on line } y = -2x + 2.5 \quad \text{--- (2)}$$



(11)

Solving ① &amp; ② simultaneously

$$2x - 2 \cdot 5 = -2x + 2 \cdot 5$$

$$4x = 5$$

$$x = \frac{5}{4}, y = 0$$

$$\therefore C \left( \frac{5}{4}, 0 \right)$$

Radius of circle is CP

$$\therefore r = \sqrt{\left(\frac{15}{4}\right)^2 + \left(\frac{15}{2}\right)^2}$$

$$= \sqrt{\frac{1125}{16}}$$

$$\therefore \text{Equation of circle} : \left(x - \frac{5}{4}\right)^2 + y^2 = \frac{1125}{16}$$

$$c) i) xy = c^2$$

$$\text{differentiating, } y + x \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{y}{x}$$

$$\text{At } T \left( ct, \frac{c}{t} \right)$$

$$m_{\text{tangent}} = -\frac{1}{t^2}$$

$$E_{\text{tangent}} : y - \frac{c}{t} = -\frac{1}{t^2} (x - ct)$$

$$t^2 y - ct = x - ct$$

$$x + t^2 y = 2ct$$

$$ii) x \text{ int} \Rightarrow y = 0$$

$$y \text{ int} \Rightarrow x = 0$$

$$\therefore x = 2ct$$

$$t^2 y = 2ct$$

$$\therefore A(2ct, 0)$$

$$y = \frac{2c}{t}$$

$$\text{Midpoint of } AB = \left( ct, \frac{c}{t} \right) = T$$

$$B \left( 0, \frac{2c}{t} \right)$$

$\angle BOA = 90^\circ \therefore AB$  is the diameter of a circle passing through O. T is the midpoint of this diameter and therefore the centre of the circle passing through O, A and B.



## Question 14

$$a) \quad i) \quad \int_0^a f(x) dx = \int_0^a f(a-x) dx$$

$$\text{RHS} = \int_0^a f(a-x) dx$$

$$\text{let } u = a - x$$

$$du = -dx$$

$$= - \int_a^0 f(u) du$$

$$\text{when } x=0, u=a$$

$$x=a, u=0$$

$$= \int_0^a f(u) du$$

$$= \int_0^a f(x) dx$$

$$= \text{LHS}$$

$$ii) \quad \int_0^2 x(2-x)^5 dx$$

$$= \int_0^2 (2-x) x^5 dx$$

(from (i))

$$= \int_0^2 (2x^5 - x^6) dx$$

$$= \left[ \frac{2x^6}{6} - \frac{x^7}{7} \right]_0^2$$

$$= \frac{64}{3} - \frac{128}{7}$$

$$= \frac{64}{21}$$



(13)

$$b) i) I_{2n+1} = \int_0^1 x^{2n+1} e^{x^2} dx$$

$$= \int_0^1 x^{2n} \cdot x e^{x^2} dx$$

$$= \left[ \frac{1}{2} x^{2n} e^{x^2} \right]_0^1 - \int_0^1 \frac{1}{2} e^{x^2} \cdot 2n x^{2n-1} dx$$

$$= \frac{1}{2} e^1 - n \int_0^1 x^{2n-1} e^{x^2} dx$$

$$I_{2n+1} = \frac{1}{2} e - n I_{2n-1}$$

$$ii) I_5 = \frac{1}{2} e - 2 I_3$$

$$= \frac{1}{2} e - 2 \left( \frac{1}{2} e - I_1 \right)$$

$$= \frac{1}{2} e - e + 2 \int_0^1 x e^{x^2} dx$$

$$= -\frac{e}{2} + \left[ e^{x^2} \right]_0^1$$

$$= -\frac{e}{2} + e^1 - e^0$$

$$= \frac{e}{2} - 1$$

$$c) i) (\cos \theta + i \sin \theta)^3$$

$$= \cos^3 \theta + 3 \cos^2 \theta (i \sin \theta) + 3 \cos \theta (i \sin \theta)^2 + (i \sin \theta)^3$$

$$= \cos^3 \theta + 3i \cos^2 \theta \sin \theta - 3 \cos \theta \sin^2 \theta - i \sin^3 \theta$$

ii) Using De Moivre's Thm,

$$(\cos \theta + i \sin \theta)^3 = \cos 3\theta + i \sin 3\theta$$

Equating real parts,

$$\cos 3\theta = \cos^3 \theta - 3 \cos \theta \sin^2 \theta$$

$$= \cos^3 \theta - 3 \cos \theta (1 - \cos^2 \theta)$$



(14)

$$= \cos^3 \theta - 3 \cos \theta + 3 \cos^3 \theta$$

$$\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$$

$$4 \cos^3 \theta = \cos 3\theta + 3 \cos \theta$$

$$\cos^3 \theta = \frac{1}{4} \cos 3\theta + \frac{3}{4} \cos \theta$$

$$\text{iii) } 4 \cos^3 \theta - 3 \cos \theta = 1$$

$$\therefore \cos 3\theta = 1 \quad (\text{from ci})$$

$$3\theta = 2k\pi$$

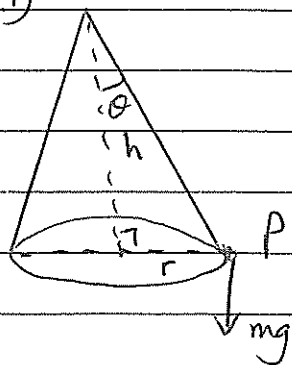
$$\theta = \frac{2k\pi}{3}$$

Smallest value occurs when  $k=1$

$$\therefore \theta = \frac{2\pi}{3}$$

Question 15

a) i)



Resolving forces vertically & horizontally

$$T \cos \theta = mg \quad \text{--- (1)}$$

$$T \sin \theta = mr\omega^2 \quad \text{--- (2)}$$

$$(2) \div (1)$$

$$\frac{T \sin \theta}{T \cos \theta} = \frac{mr\omega^2}{mg}$$

$$\tan \theta = \frac{r\omega^2}{g}$$

$$\tan \theta = \frac{r}{h}$$

$$\therefore \frac{r\omega^2}{g} = \frac{r}{h}$$

$$\omega^2 = \frac{g}{h}$$

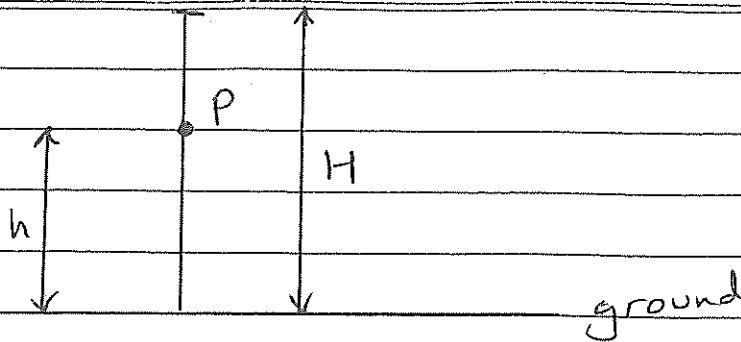
$$\omega = \sqrt{\frac{g}{h}}$$

$$\text{ii) Period} = \frac{2\pi}{\omega} = \frac{2\pi}{\sqrt{\frac{g}{h}}} = \frac{2\pi\sqrt{h}}{\sqrt{g}}$$

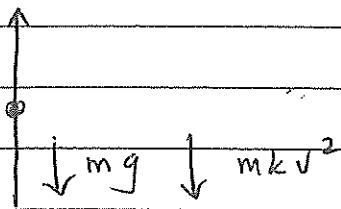


(15)

b)



i)



$$m\ddot{x} = -mg - mkv^2$$

$$\ddot{x} = -(g + kv^2)$$

$$v \frac{dv}{dx} = -(g + kv^2)$$

$$dx = \frac{-v}{g + kv^2} dv$$

$$\text{at } x = H, v = 0$$

$$x = 0, v = u$$

$$\therefore \int_0^H dx = -\frac{1}{2k} \int_u^0 \frac{2kv}{g + kv^2} dv$$

$$H = \frac{1}{2k} \int_0^u \frac{2kv}{g + kv^2} dv$$

$$= \frac{1}{2k} \left[ \ln(g + kv^2) \right]_0^u$$

$$= \frac{1}{2k} \left( \ln(g + ku^2) - \ln g \right)$$

$$H = \frac{1}{2k} \ln \left( \frac{g + ku^2}{g} \right)$$



(16)

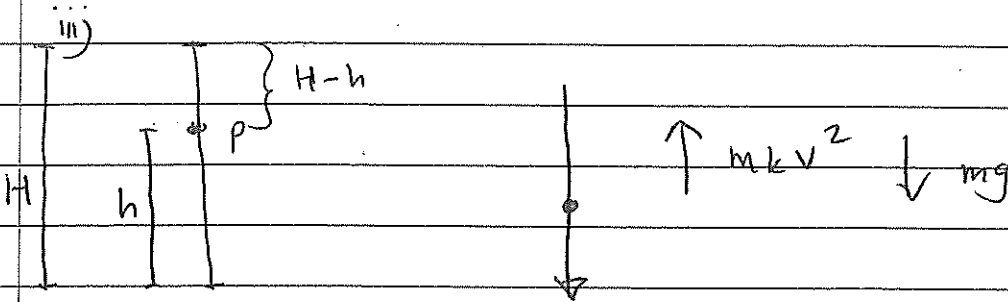
ii) when  $x=h$ ,  $v=V$ 

$$\int_0^h dx = -\frac{1}{2k} \int_u^V \frac{2kv}{g+kv^2} dv \quad (\text{from (i)})$$

$$h = \frac{1}{2k} \left[ \ln(g+kv^2) \right]_V^u$$

$$= \frac{1}{2k} \left[ \ln(g+ku^2) - \ln(g+kV^2) \right]$$

$$= \frac{1}{2k} \ln \left( \frac{g+ku^2}{g+kV^2} \right)$$



$$m\ddot{x} = mg - mkv^2$$

$$\ddot{x} = g - kv^2$$

$$v \frac{dv}{dx} = g - kv^2$$

$$dx = \frac{v}{g - kv^2} dv$$

$$\text{at } x=0, v=0$$

$$x=H-h, v=\frac{1}{2}V$$

$$\int_0^{H-h} dx = -\frac{1}{2k} \int_0^{\frac{V}{2}} \left( \frac{-2kv}{g - kv^2} \right) dv$$

$$H-h = -\frac{1}{2k} \left[ \ln(g - kv^2) \right]_0^{\frac{V}{2}}$$

$$= -\frac{1}{2k} \left[ \ln \left( g - k \frac{V^2}{4} \right) - \ln g \right] = -\frac{1}{2k} \ln \left( \frac{g - \frac{kV^2}{4}}{g} \right)$$

$$H-h = -\frac{1}{2k} \ln \left( \frac{4g - kV^2}{4g} \right) \quad \text{--- (1)}$$



(17)

Also, using  $H$  from (i) and  $h$  from (ii)

$$H-h = \frac{1}{2k} \ln \left( \frac{g+ku^2}{g} \right) - \frac{1}{2k} \ln \left( \frac{g+kv^2}{g} \right)$$

$$H-h = \frac{1}{2k} \ln \left( \frac{g+kv^2}{g} \right) \quad \text{--- (2)}$$

From (1) & (2)

$$\ln \left( \frac{g+kv^2}{g} \right) = -\ln \left( \frac{4g-kv^2}{4g} \right) = \ln \left( \frac{4g-kv^2}{4g} \right) = 1$$

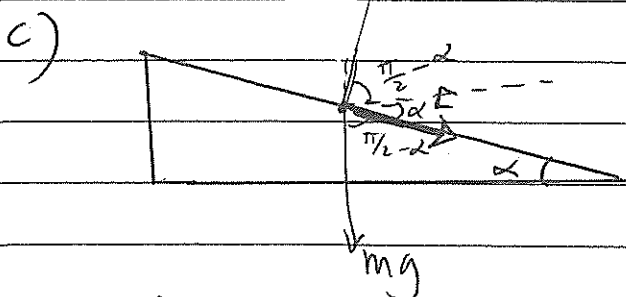
$$\frac{g+kv^2}{g} = \frac{4g}{4g-kv^2}$$

$$4g^2 + 4gkv^2 - gkv^2 - k^2v^4 = 4g^2$$

$$3gkv^2 = k^2v^4$$

$$v^2 = \frac{3g}{k}$$

$$v = \sqrt{\frac{3g}{k}}$$



Resolving Forces

Horizontally

$$\frac{mv^2}{r} = F \cos \alpha + N \cos \left( \frac{\pi}{2} - \alpha \right)$$

$$= F \cos \alpha + N \sin \alpha$$

$$\therefore F \cos \alpha = \frac{mv^2}{r} - N \sin \alpha \quad \text{--- (1)}$$

Vertically

$$N \cos \alpha - mg - F \cos \left( \frac{\pi}{2} - \alpha \right) = 0$$

$$N \cos \alpha - mg - F \sin \alpha = 0$$

$$\therefore F \sin \alpha = N \cos \alpha - mg \quad \text{--- (2)}$$



(18)

$$\text{ii) } ① \times \cos \alpha + ② \times \sin \alpha$$

$$F \cos^2 \alpha = \frac{mv^2}{r} \cos \alpha - N \sin \alpha \cos \alpha$$

$$+$$

$$F \sin^2 \alpha = N \sin \alpha \cos \alpha - mg \sin \alpha$$

$$= F (\cos^2 \alpha + \sin^2 \alpha) = \frac{mv^2}{r} \cos \alpha - mg \sin \alpha$$

$$F = \frac{m}{r} \cos \alpha \left( v^2 - \frac{gr \sin \alpha}{\cos \alpha} \right)$$

$$= \frac{m (v^2 - gr \tan \alpha)}{r} \cos \alpha$$

$$\text{iii) } r = 200 \text{ m}, g = 9.8 \text{ m/s}^2, F = 0$$

$$v = 100 \text{ km/h} = \frac{100 \times 1000}{3600} = \frac{250}{9} \text{ m/s}$$

$$\text{From (ii) } 0 = \frac{m (v^2 - gr \tan \alpha)}{r} \cos \alpha$$

$$v^2 - gr \tan \alpha = 0$$

$$\tan \alpha = \frac{v^2}{gr}$$

$$= \frac{(250/9)^2}{9.8 \times 200}$$

$$= 21.486$$

$$\alpha \approx 21^\circ 29'$$



(19)

## Question 16

$$a) \quad P(x) = x^3(ax^2 + bx + c) = ax^5 + bx^4 + cx^3$$

$$\text{Let } Q(x) = P(x) - 1 = ax^5 + bx^4 + cx^3 - 1$$

Since  $Q(x)$  is divisible by  $(x-1)^3$ ,  $x$  is a triple root

$$Q'(x) = 5ax^4 + 4bx^3 + 3cx^2$$

$$Q'(1) = 5a + 4b + 3c = 0 \quad \text{--- (1)}$$

$$Q''(x) = 20ax^3 + 12bx^2 + 6cx$$

$$Q''(1) = 20a + 12b + 6c = 0 \quad \text{--- (2)}$$

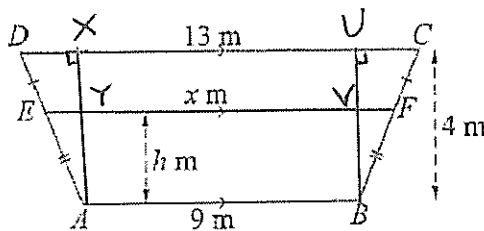
$$Q(1) = a + b + c = 0 \quad \text{--- (3)}$$

Solving (1), (2) & (3) simultaneously,

$$a = 6, \quad b = -15, \quad c = 10$$

$$\therefore P(x) = 6x^5 - 15x^4 + 10x^3$$

b)



Draw  $AX$  and  $BU \perp DC$

$$\triangle AXD \equiv \triangle BUC \text{ (RHS)}$$

$$DX = CU = 2$$

$$\frac{EY}{AY} = \frac{DX}{AX} \quad \text{(ratio of matching sides of } \equiv \Delta s)$$

$$\frac{EY}{h} = \frac{2}{4}$$

$$EY = \frac{h}{2} \quad ; \quad \text{Similarly, } VF = \frac{h}{2}$$

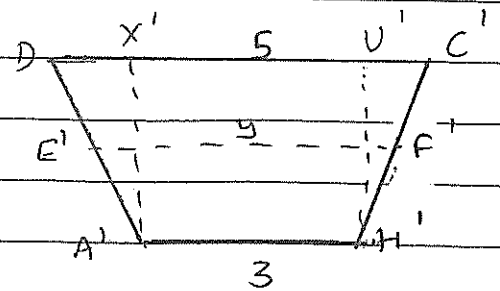
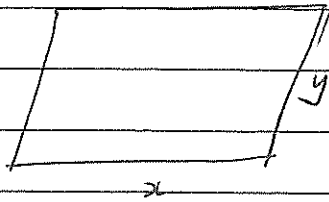
$$EF = EY + 9 + VF$$

$$x = 9 + h$$



(20)

ii) Cross-sections // to the base will be rectangles



From (i)  $x = 9 + h$

$$D'X' = C'U' = 1$$

$$\begin{aligned} \text{Area of cross-section} \\ = (9+h) \left(3 + \frac{h}{2}\right) \end{aligned}$$

$$= \frac{h^2}{2} + \frac{15h}{2} + 27$$

Using III  $\Delta$ s

$$\frac{E'Y'}{h} = \frac{1}{4}$$

$$E'Y' = \frac{h}{4}$$

$$y = 3 + 2 \times \frac{h}{4}$$

$$y = 3 + \frac{h}{2}$$

$$\Delta V = \lim_{h \rightarrow 0} \sum_{h=0}^4 \left( \frac{h^2}{2} + \frac{15h}{2} + 27 \right) \Delta h$$

$$V = \int_0^4 \left( \frac{h^2}{2} + \frac{15h}{2} + 27 \right) dh$$

$$= \left[ \frac{h^3}{6} + \frac{15h^2}{4} + 27h \right]_0^4$$

$$V = 178 \frac{2}{3} \text{ m}^3$$

(21)

c) i)  $y = mx + k$  — (1)

$xy = c^2$  — (2)

Solving (1) & (2)

$$x(mx + k) = c^2$$

$$mx^2 + kx = c^2$$

$$mx^2 + kx - c^2 = 0$$

$$\Delta = k^2 - 4mc^2$$

Since  $y = mx + k$  is tangent  $\Delta = 0$

$$\therefore k^2 - 4mc^2 = 0$$

ii) The equation of the line through  $(-1, -3)$  with gradient  $m$  is:

$$y + 3 = m(x + 1)$$

$$y = mx + m - 3$$

$$y = mx + k \quad \text{where } k = m - 3.$$

This is tangent to  $xy = 4$  if  $k^2 + 16m = 0$

$$(m - 3)^2 + 16m = 0$$

$$m^2 + 10m + 9 = 0$$

$$(m + 9)(m + 1) = 0$$

$$m = -9 \text{ or } m = -1$$

Equation of tangent  $y = mx + m - 3$

$$m = -9; \quad y = -9x - 12$$

$$m = -1; \quad y = -x - 4$$

Solving  $y = -9x - 12$  and  $xy = 4 \Rightarrow x(-9x - 12) = 4$

Solving  $y = -x - 4$  and  $xy = 4$ ,

$$x(-x - 4) = 4$$

$$x^2 + 4x + 4 = 0$$

$$(x + 2)^2 = 0$$

$$x = -2, \quad y = -2 \Rightarrow (-2, -2)$$

$$9x^2 + 12x + 4 = 0$$

$$(3x + 2)^2 = 0$$

$$x = -\frac{2}{3}, \quad y = -6$$

$$\left(-\frac{2}{3}, -6\right)$$